# BYD Deep-Dive: E/E Architecture, Software Stack & AI Strategy

Technical & Strategic Analysis 2024-2030

June 2026

Public Data

EXECUTIVE OVERVIEW

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# E/E Architecture Evolution

From 80 distributed ECUs to Xuanji central compute — BYD's hardware consolidation trajectory and zonal architecture roadmap through 2030

# BYD's E/E architecture compressed from ~80 ECUs to ~15, targeting 3 by 2030

PRE-2020

**Distributed ECUs**

**~80** ECUs

CAN backbone

Function-specific controllers

→

2021-2023

**Domain Control**

**~30** ECUs

4 domain controllers

CAN-FD + Ethernet

→

2024-2025

**Xuanji 4.0**

**~15** ECUs

Central + Zonal

Dual GbE ring backbone

→

2026-2030

**Full Central**

**~3** ECUs

Zonal + Cloud-native

AI-defined vehicle

## Key Performance Gains by Generation

| Metric | Pre-2020 | e-Platform 3.0 | Xuanji 4.0 | Xuanji 5.0 (proj.) |
|---|---|---|---|---|
| ECU Count | ~80 | ~30 | ~15 | ~7 |
| Backbone Network | CAN | CAN-FD + Eth | Dual GbE Ring | TSN Ethernet |
| Key Innovation | Function-specific | 4-domain ctrl + BYD OS | Central + zonal | Cockpit-drive integration |

Source: BYD Dream Day 2024, Company announcements, Industry analysis

# Xuanji 4.0 deploys "One Brain, Two Ends, Three Networks, Four Chains"

## ADAS Compute Tiers by Vehicle Class

| Tier | DiPilot 100 | DiPilot 300 | DiPilot 600 |
|---|---|---|---|
| Silicon | Horizon J5 / Orin N | NVIDIA Orin X | Dual Orin X |
| Compute | ~100 TOPS | 254 TOPS | 508 TOPS |
| Sensors | 12C+5R+12U | + 1 LiDAR | + 3 LiDAR |

## Xuanji Architecture Framework

**One Brain** — Central control chip (neural processing hub)

**Two Ends** — Cloud AI + Vehicle-side AI computing

**Three Networks** — Vehicle + 5G + Satellite comms

**Four Chains** — Sensing + Control + Data + Mechanical

## In-Vehicle Network Topology

- **Dual Gigabit Ethernet Ring**
  High-speed backbone for ADAS sensor data and camera feeds
- **CAN-FD** for legacy body/chassis functions
- **5G + Satellite** for cloud connectivity

- **800V Architecture** (Evo)
  Power delivery for high-performance compute modules
- **DiSus Computing Center**
  Chassis domain controller (ms-level response)

- **Zonal Gateways**
  Front/rear/left/right zone controllers aggregate local sensors/actuators
- **12-in-1 Powertrain** (Evo)
  Motor + SiC inverter + BMS integrated

Source: BYD Dream Day 2024, NVIDIA DRIVE Orin specs, Horizon Robotics

# FinDreams enables ~70% in-house sourcing — 20-25% below traditional OEM cost

VERTICAL INTEGRATION

~70%

of components sourced in-house

vs. ~30-40% for traditional OEMs

COST ADVANTAGE

20-25%

below traditional OEM baseline

for core components

2024 VOLUME

4.27M

vehicles delivered

~2.5x Tesla's volume

## FinDreams Ecosystem & In-House Integration Depth

| Subsidiary | Function | Key Products | In-House % |
|---|---|---|---|
| FinDreams Battery | Battery cells & packs | Blade Battery (LFP), Cell-to-Pack | **100%** |
| BYD Semiconductor | Semiconductors | IGBT 4.0/5.0, SiC, MCU, Sensors | **~85%** |
| FinDreams Tech | Automotive electronics | ADAS modules, cockpit, body control | **~90%** |
| FinDreams Powertrain | Powertrain systems | Motors, EHS, 8-in-1 / 12-in-1 | **~95%** |
| FinDreams Precision | Manufacturing | Molding, precision components | **~95%** |

Source: BYD Annual Report 2024, Company disclosures, Supply chain analysis

02

# Software Stack & OS Strategy

Multi-OS architecture, proprietary middleware, and the path to service-oriented vehicles

# BYD runs Linux-based BYD OS, Android DiLink cockpit, and QNX for safety domains

### BYD OS — Platform Layer

- Hardware-software decoupling
- 4-domain controller integration
- Full-vehicle OTA support
- Linux kernel + RT extensions
- Introduced with e-Platform 3.0 (2021)
- 50% response efficiency gain

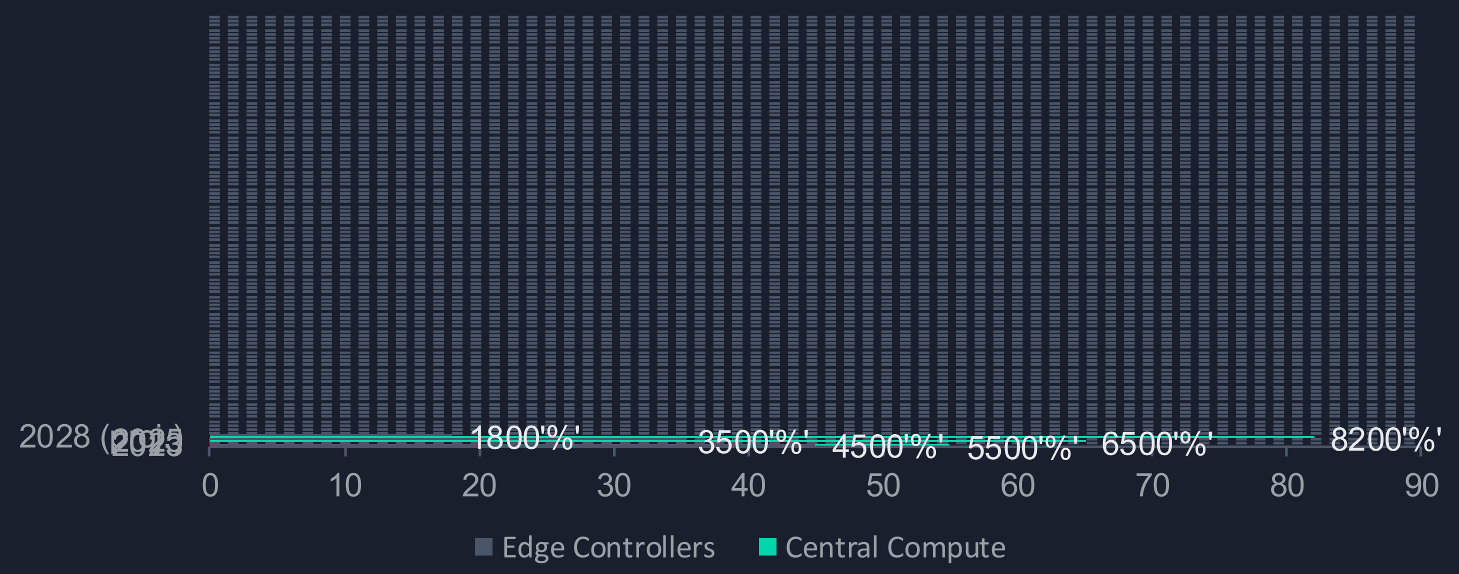
### DiLink — Cockpit (Android)

- Android AOSP-based
- Snapdragon 690 (DiLink 4.0)
- 5G + C-V2X connectivity
- Alibaba Cloud DiCloud AI
- 15.6" rotatable touchscreen
- 5 generations of evolution

### QNX — Safety-Critical

- ISO 26262 ASIL-D certified
- Microkernel architecture
- Hypervisor: Android + QNX
- Hardware-enforced isolation
- CPU/MMU/I/O virtualization
- Deterministic real-time response

## Software Running on Central Compute vs. Edge




### Virtual ECU (vECU) Roadmap

BYD is expanding virtualization to support vECUs running legacy controller functions as software tasks on centralized platforms.

- Current: Hypervisor for cockpit domain separation
- 2026-28: vECUs for body/comfort on zonal controllers
- Investing in AUTOSAR Adaptive compatibility

Source: BYD OS documentation, QNX whitepapers, Industry analysis

# Proprietary SOA middleware with selective AUTOSAR Classic for edge controllers

## Middleware & SOA Strategy

**In-house Proprietary SOA** for Xuanji "Four Chains":

- Sensing chain — sensor data fusion services
- Control chain — actuator command services
- Data chain — telemetry and AI training data
- Mechanical chain — chassis/powertrain coordination

**AUTOSAR Classic** for MCU edge controllers (powertrain, chassis)

**No full AUTOSAR Adaptive** — prefers BYD OS flexibility

**Not in SOAFEE** — independent standards posture

## Safety Certification Approach

**ISO 26262 Compliance:**

- ASIL-D: NVIDIA Orin (DiPilot 300/600)
- ASIL-B: Cockpit domain (Android limits)
- QNX Hypervisor: hw-enforced isolation

**Mixed-Criticality Management:**

- Physical separation of highest-criticality domains
- Dedicated safety controllers (DiSus, BMS)
- Independent power + watchdog for edge ctrl

## OEM Middleware & Standards Posture Comparison

| Dimension | BYD | Tesla | VW | Xiaomi |
|---|---|---|---|---|
| Middleware | Proprietary (Xuanji) | Fully proprietary | VW.OS (standardized) | HyperOS (proprietary) |
| AUTOSAR | Classic only (selective) | None | Classic + Adaptive | Limited |
| Ecosystem Openness | Closed (in-house) | Closed | Open (third-party) | Open (Android-based) |

Source: BYD technical disclosures, AUTOSAR membership data, Industry analysis

03

# ADAS, AI & Silicon Strategy

Three-tier DiPilot system, multi-source silicon strategy, and DeepSeek AI integration

# DiPilot / God's Eye covers all price tiers from $10K Seagull to $1M Yangwang

| Tier | Name | Compute | LiDAR | Vehicle Application | Capabilities |
|---|---|---|---|---|---|
| **Entry** | God's Eye C | Horizon J5 / Orin N | None | Seagull, Dolphin, ATTO 3 | Highway NOA, lane change, parking |
| **Mid** | God's Eye B | NVIDIA Orin X | 1x | Denza, Dynasty/Ocean > $20k | City + highway NOA, adv. parking |
| **Premium** | God's Eye A | Dual Orin X (508 TOPS) | 3x | Yangwang U8, U9 | Full urban autonomy, track demo |

### ADAS Organization & Investment

**5,000+**
ADAS engineers
(3,000 software, 1,000+ algorithms)

**RMB 100B**
Committed to intelligent
driving & AI R&D

**21**
Models with standard ADAS
as of Feb 2025

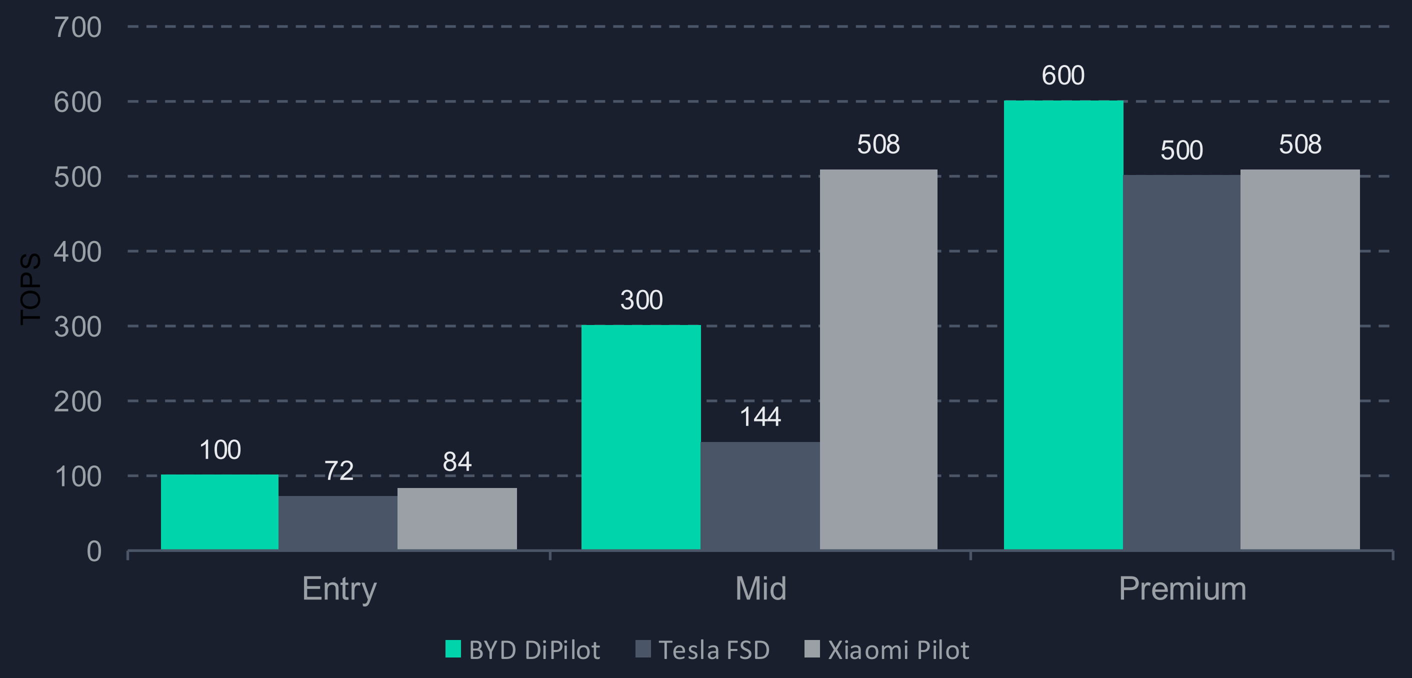
**150M km**
Daily ADAS training data
from fleet (end 2025)

**In-house + Partnered Development:** Self-developed algorithms (Tianxuan+Tianlang merged team) + Horizon Robotics (Journey chips) + Momenta (certain ADAS) + Huawei Qiankun ADS 3.0 (optional in Fangchengbao Bao 5)

Source: BYD Intelligent Strategy Conference Feb 2025, 36Kr, Company disclosures

# NVIDIA for premium, Horizon for volume, proprietary chips in long-term R&D

| Vendor | Product | TOPS | Application in BYD | Tier |
|---|---|---|---|---|
| NVIDIA | DRIVE Orin N | 84 | DiPilot 100 (some variants) | Entry |
| NVIDIA | DRIVE Orin X | 254 | DiPilot 300 (1x) / 600 (2x) | Mid / Premium |
| Horizon | Journey 5 | 128 | DiPilot 100 | Entry |
| Horizon | Journey 6M | ~128 | DiPilot 100 (next-gen) | Entry |
| BYD Semi | IGBT 4.0/5.0, SiC | N/A | Powertrain, 800V inverters | Power domain |




**Horizon Robotics Advantage:**

- BPU Nash: 85% compute utilization vs. NVIDIA 30-40%
- ~RMB 1,000 cheaper per vehicle than NVIDIA
- 137 kDMIPS CPU (J6M) > Orin N

**BYD Semiconductor:**

- IGBT 4.0: 98.5% efficiency in EHS
- Acquired Ningbo SinoMOS (TSMC Fab 1) in 2008
- Exploring proprietary ADAS chip (long-term)

Source: NVIDIA, Horizon Robotics, BYD Semiconductor disclosures

# DeepSeek R1 integration and 150M km/day training data power BYD's AI flywheel

## DeepSeek R1 Integration (Feb 2025)

- LLM adapted for real-time driving perception & decision-making
- Enhanced cloud training efficiency for ADAS data
- In-vehicle reasoning: understands vague user intent
- Cockpit personalization via implicit need detection
- Cost-effective vs. in-house LLM development

## Xuanji AI Foundation Model

- Industry's first dual-cycle multi-modal AI model
- 300+ vehicle usage scenarios covered
- Dual-cycle: vehicle (ms) + cloud (continuous learning)
- 72M km daily training data (2024)
- Largest automotive data foundation in China

## The Data Flywheel Effect

**4.27M vehicles**
on the road
→
**150M km/day**
training data
→
**DeepSeek + Xuanji**
AI model training
→
**Better ADAS**
attracts more buyers

## End-to-End Neural Pipeline

Hired Zhou Peng (ex-Baidu cabin-driving integration) in June 2024 to lead end-to-end large model development. Tianxuan+Tianlang merged team focuses on perception-to-planning neural networks with rule-based safety fallbacks. Conservative "mechanical" definition of end-to-end (sensor-to-actuator chain) with growing neural component.

Source: BYD Intelligent Strategy Conference Feb 2025, 36Kr, DeepSeek integration announcements

04

# Connectivity, OTA & Competitive Position

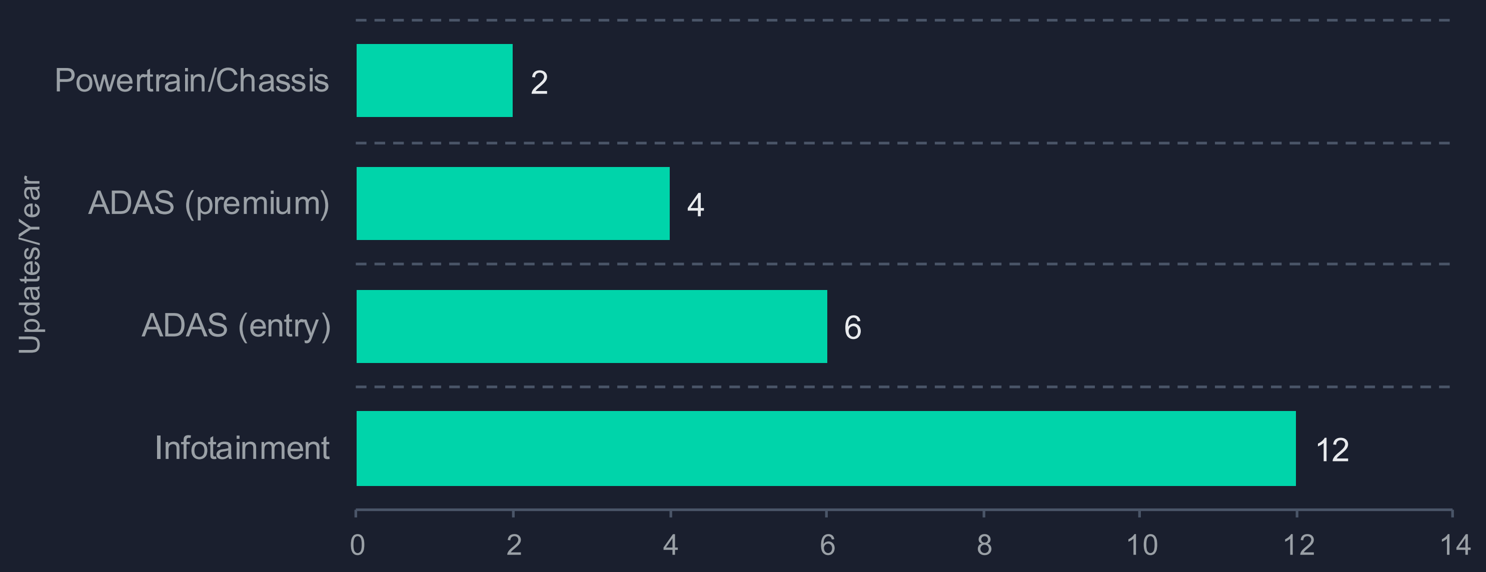
Connected-car infrastructure, organizational scale, and the SDV litmus test against Tesla and Xiaomi

# Full-vehicle OTA with dual-bank flashing, delta compression, and staged rollout

## OTA Architecture & Security

- **Dual-bank (A/B) partitioning** at central + edge
- **Delta compression** — only differences transmitted
- **Cryptographic signature verification** via HSE
- **Automatic rollback** on failure or health check fail
- **Staged rollout**: fleet subset validation first
- Cloud on **Alibaba Cloud** + **Huawei SAP HANA**

## OTA Deployment Cadence by Domain




## Ground-Up Connectivity Stack

**Xuanji "Three Networks"**

Vehicle network + 5G cellular + satellite comms (Yangwang U8 Off-road)

**C-V2X Integration**

Cooperative adaptive cruise, intersection collision warning, emergency preemption

**DiCloud AI Platform**

Alibaba Cloud backend for voice AI, OTA distribution, fleet data pipeline

Source: BYD technical documentation, Alibaba Cloud partnership, Industry analysis

# 120K+ R&D engineers and RMB 40B annual R&D — world's largest auto R&D machine

**120,000+**
R&D engineers globally
11 research institutes

**RMB 40B+**
Annual R&D spend (2024)
2x increase from 2023

**30 hrs**
Assembly time per vehicle
vs. 40-50 hrs industry avg

**150K/yr**
Per plant capacity
Thailand, Hungary, Brazil, Turkey

### Key Engineering Organizations

| Organization | Function | Headcount | Key Leader |
|---|---|---|---|
| New Auto Tech Research Inst. | Overall intelligent tech R&D | 15,000 | Yang Dongsheng |
| Self-Driving Dept. | ADAS algorithm & software | 5,000+ | Li Feng |
| Core Algorithm Team | Perception, planning, AI models | 1,000+ | — |
| BYD Semiconductor | Chip design & manufacturing | ~3,000 | — |
| Intelligent Cockpit Team | DiLink, HMI, connectivity | ~2,000 | — |

**"Horse Racing" Culture:** BYD encourages internal competition between parallel teams. Tianxuan and Tianlang departments merged in 2024 after competing on ADAS algorithms. Monthly ADAS payroll exceeds RMB 1B (~$140M). Chairman Wang Chuanfu personally tracks intelligent driving progress. "Horse racing" has accelerated innovation but also caused resource redundancy and product delays.

Source: BYD Annual Report 2024, 36Kr, Company disclosures

# BYD trails Tesla in software purity but leads in scale, cost, and vertical integration

## The SDV Litmus Test: Multi-Dimension Comparison

| Dimension | BYD | Tesla | Xiaomi |
|---|---|---|---|
| HW-SW Decoupling | BYD OS (partial) | Full (proprietary OS) | HyperOS (full) |
| OTA Depth | All domains | All domains | All domains |
| Centralized Compute | Xuanji 4.0 (hybrid) | Full central (HW4) | Domain-separated |
| Proprietary Silicon | No (NVIDIA/Horizon) | Yes (FSD chip) | No (NVIDIA/Qualcomm) |
| Data Flywheel | 150M km/day | Significant | Limited (new) |
| Vertical Integration | ~70% in-house | ~40-50% | ~20-30% |
| 2024 Volume | **4.27M vehicles** | 1.79M vehicles | ~136K vehicles |

## Strategic Assessment

**BYD's Hybrid SDV Model:**

Proprietary middleware + third-party silicon + in-house algorithms + external AI partnerships (DeepSeek, Horizon). This pragmatic approach trades software purity for manufacturing velocity and cost leadership.

**Key Gaps vs. Tesla:**

- No proprietary ADAS SoC (FSD chip)
- End-to-end neural ADAS less mature
- OTA cadence and scope narrower
- Software ecosystem less open

Source: Company reports, Industry analysis, Production data 2024

# Strategic Implications

BYD is building the world's most vertically integrated automotive technology stack. By 2028, the convergence of **Xuanji 5.0 zonal architecture**, **DeepSeek AI**, and **150M km/day of fleet data** could close the software gap with Tesla while maintaining an unassailable cost and scale advantage.

The question is not whether BYD can become software-defined — it is whether any competitor can match its **economics** while catching its **software trajectory**.

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Technical & Strategic Analysis 2024-2030 | June 2026